# ATAC-seq Pipeline

**Version 2.0.0**

## Overview

Samples were processed with the BIC ATAC-seq pipeline, which takes aligned, duplicate-marked BAM files and produces filtered alignments, normalized signal tracks (bigWig), MACS2 peak calls for each sample, a merged peak atlas with a raw fragment count matrix, QC metrics and, when the design has replicates, a differential accessibility analysis. The code is at https://github.com/soccin/ATAC-seq/releases/tag/v2.0.0.

Read filtering, the Tn5 shift and peak calling follow the ENCODE ATAC-seq pipeline (ENCODE-DCC/atac-seq-pipeline v2.2.3) with four differences, described below: the mapping quality cutoff, where duplicates are marked (by PEMapper, before the read filter), and two additional read filters. The signal tracks, the peak atlas and the count normalization follow the methods of R. Koche (MSKCC Center for Epigenetics Research).

Reads were aligned to one of three reference genomes: GRCh37 (b37), GRCh38 (the GDC `GRCh38.d1.vd1` build, hg38) or GRCm38 (mm10). The build of a project is recorded in the BAM headers. Peaks, signal tracks and the atlas use the autosomes, X and Y only.

## Methods

**Alignment.** Paired-end reads were aligned with BWA-MEM 0.7.19 (Li 2013) using the PEMapper pipeline (https://github.com/soccin/PEMapper), which trims the TruSeq adapter (`AGATCGGAAGAGC`) with cutadapt, aligns with `bwa mem -M`, merges lanes, sorts, and marks duplicates with Picard MarkDuplicates 2.25.5. The Nextera adapter of ATAC-seq libraries is not trimmed; BWA-MEM soft-clips it at the 3' end of the read. Duplicates are marked in the BAM, not removed. The PEMapper version is in the `@PG` header line of each BAM.

**Read filtering.** With samtools 1.20 (Danecek 2021) each BAM was reduced to properly paired reads with mapping quality 10 or more, without unmapped reads, reads whose mate is unmapped, secondary alignments, reads failing vendor QC or duplicates (`-q 10 -f 3 -F 1804`). The cutoff of 10 is R. Koche's; the ENCODE pipeline by default keeps reads with up to four alignments, or uses a cutoff of 30 when multimapping reads are not kept. The duplicates removed here were marked by PEMapper on all alignments, before this filter; the ENCODE pipeline marks duplicates after its own filter. Two further filters, which the ENCODE pipeline does not have, correct for an artifact of BWA-MEM local alignment. A read whose sequence is not from the reference, or only partly from it, can still be reported as a short alignment of 19 to 35 bp with the rest of the read soft-clipped. Such reads carry the same non-reference sequence, so hundreds or thousands of them pile up on single bases, mostly inside peaks; in some libraries they are about half of all read pairs. Bowtie2 in end-to-end mode, the ENCODE aligner, cannot produce these alignments, so ENCODE needs no such filter. The first filter removes every read whose alignment is soft-clipped at its 5' end, the Tn5 insertion site: a forward-strand read whose CIGAR string starts with `S` or a reverse-strand read whose CIGAR string ends with `S`. The reads were then name-sorted and passed through `samtools fixmate -r`, and the flag filter was applied again, as in the ENCODE pipeline, which removes a read

whose mate did not pass the filters above. The second filter was applied at the same step: it removes every pair with an insert of 30 bp or less (|TLEN| <= 30). The result was coordinate-sorted with Picard SortSam 3.4.0; this filtered BAM is the input to every later step.

**Tn5 shift.** Each read was converted to a BED interval with bedtools 2.31.1 (Quinlan 2010) and shifted to place its 5' end at the Tn5 insertion site: +4 bp on the forward strand and -5 bp on the reverse strand (Buenrostro 2013). Reads on the mitochondrial contig were dropped at this step, as in the ENCODE pipeline, and so were reads on unplaced, random, decoy and viral contigs, which ENCODE removes only from its final peaks; the fraction of mitochondrial reads is reported as a QC metric. This set of shifted reads is the input to the signal tracks and to peak calling.

**Signal tracks.** For each sample the coverage of the shifted reads was computed with bedtools genomecov, each read as its own interval with no extension, scaled to 10 million reads (the scale factor is 10 million divided by the number of shifted reads of the sample), and written as a bigWig with the UCSC wigToBigWig tool (Kent 2010). Tracks are therefore comparable between samples by sequencing depth; the DESeq2 scale factors below are not applied to them.

**Peak calling.** Peaks were called for each sample with MACS2 2.2.9.1 (Zhang 2008) on the shifted reads, with the settings of the ENCODE ATAC-seq pipeline: `--nomodel --shift -75 --extsize 150 --keep-dup all --call-summits -p 0.01` and the genome size `hs` or `mm`. Each read contributes a 150 bp window centered on its insertion site, every read is kept (duplicates were already removed by the filter), and peaks are called at a p-value of 0.01, a permissive cutoff chosen by ENCODE so that peaks can be ranked downstream. MACS2's own output (narrowPeak, summits, xls) is delivered unchanged. The peaks were then post-processed as ENCODE does: sorted by p-value, the 300,000 most significant kept, clipped to the contig ends, peaks overlapping the ENCODE blacklist of the genome removed (Amemiya 2019; Boyle lab blacklist v2 for hg38 and mm10; for b37 the hg19 DAC consensus excludable list that ENCODE uses, with the `chr` prefix removed) and the score column capped at 1000. These `blacklistFiltered` peaks and summits are delivered beside MACS2's files and are the peaks used for the atlas.

**Peak atlas.** The blacklist-filtered peaks of all samples were pooled and merged with bedtools merge, joining peaks within 500 bp of each other into one region (R. Koche's method; ENCODE does not merge peaks). Any blacklisted region bridged by a merge was subtracted, so no atlas region overlaps the blacklist. The result is the peak atlas, one set of regions shared by every sample of the project, written in SAF format (1-based, inclusive coordinates) with the regions named `Peak_1`, `Peak_2`, … in genome order. Because peaks are merged and pooled over samples, the number of atlas regions grows with the number of samples and is not comparable to ENCODE's per-sample peak-count standards; the per-sample `narrowPeak` files are.

**Counts.** Fragments were counted over the atlas with featureCounts 2.1.1 (Liao 2014) from the filtered BAMs, counting read pairs as single fragments (`-p --countReadPairs`), with a fragment that overlaps two atlas regions counted in both (`-O`), and with the same mapping quality cutoff as the read filter (`-Q 10`). The counts were verified by two independent checks before use: each sample's fragment total must be half its mapped reads, within 5%, and the number of fragments in the atlas, counted a second time with samtools alone, must equal featureCounts' assigned total exactly. A run whose counts fail either check is stopped.

**Normalization.** Size factors were computed from the raw count matrix with DESeq2 (Love 2014; median-of-ratios method, Anders 2010) and written as scale factors, the reciprocal of the DESeq2 size factor, so that a normalized count is the raw count multiplied by the sample's scale factor. They

are not applied to any delivered file and are not part of the standard delivery; they are available on request.

**QC.** The insert-size distribution of the filtered BAM was computed with Picard CollectInsertSizeMetrics over a fixed range of 0 to 1000 bp, as in the ENCODE pipeline, and plotted for all samples, to show the nucleosome-free and nucleosomal fragment populations. Its smallest insert is 31 bp, because of the insert filter above. The mitochondrial fraction is the number of reads on the mitochondrial contig divided by all reads of the filtered BAM, so it is computed after duplicate removal; ENCODE computes its fraction on all alignments before filtering, and because mitochondrial reads are highly duplicated the fraction here is the lower of the two. TSS enrichment was computed with an R port of ENCODE's `encode_task_tss_enrich.py` on the filtered BAM: the coverage of the reads, each moved by half the read length so that it is centered on its 5' end, in a window of 2 kb on each side of every unique protein-coding TSS, averaged over TSSs in 400 bins and divided by the mean coverage of the outer 100 bp on each side; the score is the maximum of this profile. The TSS sets are the GENCODE v24 (hg38), v19 (b37) and vM7 (mm10) files of ENCODE's first genome data release, not the newer files the current ENCODE pipeline uses, and ENCODE states its cutoffs for GRCh38 and mm10 on RefSeq TSSs, so the cutoffs given with the output are a guide. The fraction of fragments in peaks is the number of fragments featureCounts assigned to the atlas divided by all fragments of the filtered BAM, taken from the featureCounts summary. It measures the same thing as ENCODE's FRiP, signal against background, but is a different quantity: the regions are the merged atlas of all samples rather than the sample's own peaks, and the denominator includes mitochondrial fragments, so it is not comparable to ENCODE's FRiP cutoffs. It and a principal component analysis of the samples (edgeR TMM-normalized log2 CPM over the atlas, after filterByExpr) were plotted from the count matrix, with each sample's fragments in the atlas (its depth). To tell an effect of depth from one of the data, the PCA was repeated after the counts of every sample were thinned at random (binomial, fixed seed) to about the depth of the smallest.

**Differential accessibility.** When the design has replicates, pairwise comparisons between groups were tested with edgeR (Robinson 2010; McCarthy 2012). For each comparison, the samples of its two groups were taken from the count matrix, atlas regions with too few counts removed with filterByExpr (over all samples, then over the samples of the comparison), library sizes normalized with TMM, dispersions estimated, and a negative binomial generalized linear model fitted and tested with the likelihood ratio test. P-values were adjusted with the Benjamini-Hochberg method (Benjamini 1995) and regions with FDR $< 0.05$ were called significant. The sign of the log fold change is group 2 minus group 1 of the comparison: a positive value means more accessible in group 2. Significant regions were annotated with the nearest gene and genomic feature using ChIPseeker (Yu 2015) with the UCSC knownGene annotation of the build and a TSS region of 5 kb on each side. For each comparison, every tested region was plotted as an MA plot (log fold change against mean log2 CPM) and a volcano plot (log fold change against p-value, on a log scale), with the significant regions colored by the group they are more accessible in.

**Pathway enrichment.** When requested, each comparison was tested for pathway enrichment by preranked gene set enrichment analysis (Subramanian 2005) with fgsea (Korotkevich 2021). Every tested region was assigned to the gene with the nearest TSS with ChIPseeker and the UCSC knownGene annotation of the build, and regions farther from that TSS than a set distance (by default 1 kb on either side, the promoter) were dropped. Each region was scored as the sign of its log fold change times -log10 of its p-value, and each gene took the score of its region with the largest absolute score. The genes ranked by this score were tested against MSigDB (Liberzon 2015) gene sets of 15 to 500 ranked genes, from msigdbr, with fgseaMultilevel (no lower bound on the p-value,

fixed seed); p-values were adjusted with the Benjamini-Hochberg method within each collection of each comparison. Human data used the human MSigDB collections, mouse data the mouse MSigDB (Castanza 2023). A positive normalized enrichment score means the set's genes are more accessible in group 2.

## Tool versions

| Tool | Version |
|---|---|
| BWA-MEM (PEMapper) | 0.7.19 |
| Picard MarkDuplicates (PEMapper) | 2.25.5 |
| samtools | 1.20 |
| Picard SortSam, CollectInsertSizeMetrics | 3.4.0 |
| bedtools | 2.31.1 |
| UCSC bedClip | v369 |
| UCSC wigToBigWig | kent source v361 |
| MACS2 | 2.2.9.1 |
| featureCounts (subread) | 2.1.1 |
| R | 4.5.1 |
| DESeq2 | 1.50.2 |
| edgeR | 4.8.2 |
| ChIPseeker | 1.46.1 |
| TxDb.Hsapiens.UCSC.hg38.knownGene (hg38) | 3.22.0 |
| TxDb.Hsapiens.UCSC.hg19.knownGene (b37) | 3.22.1 |
| TxDb.Mmusculus.UCSC.mm10.knownGene (mm10) | 3.10.0 |
| fgsea | 1.36.2 |
| msigdbr | 26.1.1 |
| MSigDB | 2026.1 (Hs, Mm) |
| ENCODE ATAC-seq pipeline (settings followed) | 2.2.3 |

## References

Amemiya HM, Kundaje A, Boyle AP. The ENCODE Blacklist: identification of problematic regions of the genome. Sci Rep 2019;9:9354. doi:10.1038/s41598-019-45839-z

Anders S, Huber W. Differential expression analysis for sequence count data. Genome Biol 2010;11:R106. doi:10.1186/gb-2010-11-10-r106

Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J R Stat Soc B 1995;57:289-300.

Buenrostro JD, Giresi PG, Zaba LC, Chang HY, Greenleaf WJ. Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position. Nat Methods 2013;10:1213-1218. doi:10.1038/nmeth.2688

Castanza AS, Recla JM, Eby D, Thorvaldsdottir H, Bult CJ, Mesirov JP. Extending support for mouse data in the Molecular Signatures Database (MSigDB). Nat Methods 2023;20:1619-1620. doi:10.1038/s41592-023-02014-7

Danecek P, et al. Twelve years of SAMtools and BCFtools. GigaScience 2021;10:giab008. doi:10.1093/gigascience/giab008

ENCODE ATAC-seq pipeline. ENCODE Data Coordination Center. https://github.com/ENCODE-DCC/atac-seq-pipeline. Data standards: https://www.encodeproject.org/atac-seq/

Kent WJ, Zweig AS, Barber G, Hinrichs AS, Karolchik D. BigWig and BigBed: enabling browsing of large distributed datasets. Bioinformatics 2010;26:2204-2207. doi:10.1093/bioinformatics/btq351

Korotkevich G, Sukhov V, Budin N, Shpak B, Artyomov MN, Sergushichev A. Fast gene set enrichment analysis. bioRxiv 2021;060012. doi:10.1101/060012

Li H. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. arXiv 2013;1303.3997.

Liao Y, Smyth GK, Shi W. featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. Bioinformatics 2014;30:923-930. doi:10.1093/bioinformatics/btt656

Liberzon A, Birger C, Thorvaldsdottir H, Ghandi M, Mesirov JP, Tamayo P. The Molecular Signatures Database (MSigDB) hallmark gene set collection. Cell Syst 2015;1:417-425. doi:10.1016/j.cels.2015.12.004

Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biol 2014;15:550. doi:10.1186/s13059-014-0550-8

McCarthy DJ, Chen Y, Smyth GK. Differential expression analysis of multifactor RNA-Seq experiments with respect to biological variation. Nucleic Acids Res 2012;40:4288-4297. doi:10.1093/nar/gks042

Picard Toolkit. Broad Institute. http://broadinstitute.github.io/picard/

Quinlan AR, Hall IM. BEDTools: a flexible suite of utilities for comparing genomic features. Bioinformatics 2010;26:841-842. doi:10.1093/bioinformatics/btq033

Robinson MD, McCarthy DJ, Smyth GK. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. Bioinformatics 2010;26:139-140. doi:10.1093/bioinformatics/btp616

Subramanian A, Tamayo P, Mootha VK, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. Proc Natl Acad Sci USA 2005;102:15545-15550. doi:10.1073/pnas.0506580102

Yu G, Wang LG, He QY. ChIPseeker: an R/Bioconductor package for ChIP peak annotation, comparison and visualization. Bioinformatics 2015;31:2382-2383. doi:10.1093/bioinformatics/btv145

Zhang Y, et al. Model-based analysis of ChIP-Seq (MACS). Genome Biol 2008;9:R137. doi:10.1186/gb-2008-9-9-r137